

BREAK INTO AEROSPACE

Rocket Propulsion Engineering

Subdomain Interview Preparation Guide

Targeted for the 2025–2026 Recruitment Cycle

Rocket propulsion engineering is the discipline that determines whether a launch vehicle reaches orbit, whether a spacecraft can execute a mission-critical burn, and whether a missile achieves its intended trajectory. It is one of the most demanding engineering fields that exists — sitting at the intersection of thermodynamics, fluid mechanics, combustion chemistry, structural mechanics, heat transfer, control theory, and materials science — and it is one of the few fields where the consequence of a design error is typically the complete destruction of the system in a fraction of a second, often on live television.

The field is also undergoing its most significant structural transformation since the Space Shuttle era. The emergence of SpaceX as a vertically integrated manufacturer has compressed development timescales and unit costs in ways that have forced every incumbent to respond. Rocket Lab, Relativity Space, Ursa Major, Launcher, Agnikul Cosmos, Orbex, Isar Aerospace, Rocket Factory Augsburg, HyImpulse, and dozens of other new entrants are building engines and vehicles simultaneously. The legacy primes — Aerojet Rocketdyne (now L3Harris), Northrop Grumman, Safran, ArianeGroup, JAXA/IHI, Mitsubishi Heavy Industries, RD AMROSS — continue to produce the engines that underpin established launch programmes. NASA and ESA fund advanced propulsion research at JPL, MSFC, Glenn Research Center, and ESTEC. The technical baseline is shared across all of these; the development methodology, manufacturing philosophy, test culture, and interview format differ substantially.

This document is structured as an interview preparation guide for engineers at all levels targeting roles in liquid propellant engine design, solid rocket motor development, combustion and propellant chemistry, turbopump and fluid systems engineering, nozzle and thermal protection design, test operations, or propulsion systems integration.

PART I · WHAT ROCKET PROPULSION ENGINEERING ACTUALLY LOOKS FOR

By Level

- **Intern / Graduate** — At entry level, propulsion employers are not primarily screening for engine-specific knowledge — they are screening for thermodynamic rigour, comfort with combustion chemistry, and the ability to work with the performance equations that describe what an engine does without being told which equation applies. The question that most reliably distinguishes strong from weak graduate candidates is not “what is specific impulse?” but “if I increase my chamber pressure by a factor of two, what happens to specific impulse, thrust, and the required pump power, and what new failure modes become relevant?” — a question that requires understanding the Rocket Equation, the thrust coefficient, the characteristic velocity, and the relationship between chamber pressure and turbopump work simultaneously.

Graduate interviews at legacy primes and government facilities — Aerojet Rocketdyne, Northrop Grumman Space Systems, Safran Aircraft Engines, NASA MSFC, DLR Lampoldshausen — typically involve a structured technical screen covering thermodynamics, fluid mechanics, and basic combustion, followed by one or two interviews mixing domain conceptual questions with behavioural assessment. At new-space companies — SpaceX, Rocket Lab, Ursa Major, Orbex — the technical screen tends to be faster and

more problem-oriented: a take-home or live problem involving a simplified engine sizing calculation or a combustion stability analysis. The new-space companies specifically assess whether a graduate can make reasonable engineering judgements quickly rather than waiting for full analysis.

What distinguishes graduate candidates in propulsion is evidence of genuine engagement with the discipline beyond coursework. A thesis involving combustion simulation, propellant characterisation, or nozzle design; participation in a student rocketry team (ESRA Spaceport America Cup, Intercollegiate Rocket Engineering Competition, a university liquid engine project); or a personal project involving a small-scale engine test — even a cold-flow test — is immediately differentiating. Propulsion is a field where the gap between someone who has watched combustion happen and someone who has only read about it is measurable in the first five minutes of a technical interview.

- **Mid-Level (Engineer, Senior Engineer, ~3–8 years)** — At mid-level, propulsion interviews become system-specific and failure-mode oriented. The interviewer is not assessing whether you understand the Rocket Equation — they are assessing whether you can diagnose why a turbopump seal is failing, design a regenerative cooling channel that does not burn through, or identify the coupling mechanism responsible for a low-frequency combustion oscillation observed in a recent hot fire. The questions are grounded in real hardware behaviour, not idealised theory.

Technical interviews at mid-level commonly include whiteboard problems on engine cycle analysis, heat transfer in a combustion chamber or nozzle, turbopump performance estimation, or combustion stability characterisation. Companies with a strong test focus — SpaceX Raptor team, Rocket Lab Rutherford and Archimedes, Ursa Major, Aerojet Rocketdyne — ask candidates to describe their test data interpretation process: how do you distinguish an instrumentation anomaly from a real combustion event? What does a pressure trace tell you about combustion stability, and what does it not tell you? The mid-level propulsion interview is often, at its core, a test of whether the candidate has spent time in front of test data.

The specific context driving mid-level hiring in 2025–2026: SpaceX’s Raptor 3 development and Starship’s continued test programme; United Launch Alliance’s Vulcan Centaur entering service with BE-4 engines; Rocket Lab’s Archimedes engine development for Neutron; ArianeGroup’s Prometheus reusable engine development for Ariane NEXT; Isar Aerospace, RFA, and Orbex simultaneously developing first-stage engines in Europe; NASA’s continued SLS RS-25 production and advanced propulsion research at MSFC; and a significant expansion in hypersonic propulsion research across the US, UK, Australia, and India creating demand for scramjet and dual-mode ramjet engineers. Engineers who understand the full envelope — from ignition through steady-state operation through shutdown — of a real bipropellant engine are in persistent demand across all of these simultaneously.

- **Senior (Principal Engineer, Technical Lead, Chief Engineer)** — At senior level, propulsion employers are looking for engineers who can own the technical architecture of a complete propulsion system — from propellant selection through engine cycle selection through component sizing through test campaign design through flight qualification — and who can make defensible architectural choices when the requirements are competing and the answer requires integrating combustion physics, structural mechanics, and operational constraints simultaneously.

The senior propulsion interview is a peer discussion with technical leadership, often structured around the candidate’s most significant past contribution: what was the engine, what was the engineering problem you personally owned, what was the trade you made, and what did you learn when the hardware told you you were wrong? The expectation at this level is not perfect foresight — it is the ability to learn from hardware failure faster than the competition, and to build test campaigns that force the hardware to reveal its failure modes before flight.

Senior roles at prime contractors, new-space operators, and government agencies require engineers who have taken a propulsion system from requirements through test and, ideally, through flight. The flight heritage credential is specifically valued in this field in a way that it is in almost no other engineering discipline — because the number of propulsion systems that have actually flown is small, and the engineers who have taken one there have learned things that cannot be learned any other way.

Hands-On Experience and Domain Knowledge Treated as Strong Signals

Domain

Engine Performance Analysis

How Rocket Propulsion Interviews Differ from Generic Aerospace Roles

- **The physics is multidisciplinary and must be held simultaneously.** Rocket propulsion requires thermodynamics, fluid mechanics, combustion chemistry, heat transfer, structural mechanics, and dynamics. In propulsion, a single engineer may be responsible for all of them simultaneously. A candidate who is strong in thermodynamics but cannot estimate the wall heat flux in a combustion chamber has a visible gap.
- **Test data interpretation is a core interview competency.** Employers evaluate candidates specifically on their ability to interpret test data: pressure traces, mass flow data, and vibration spectra. Identifying the ignition delay, thrust buildup, steady-state, shutdown transient, and anomalous oscillations is a practical daily skill.
- **Failure mode knowledge is tested explicitly.** Rocket propulsion interviews commonly probe failure mode knowledge directly: “what are the three most likely failure modes during the startup transient, and what does each look like in test data?” Engineers who have studied specific propulsion failure cases (F-1 instability, SSME turbopump failures) demonstrate highly valued engaged curiosity.
- **New-space interviews move faster and expect faster decisions.** SpaceX, Rocket Lab, Ursa Major, and comparable new-space companies interview with an expectation of decisiveness. The question is “what is your answer, now, with the information available, and how would you validate it?” Hedging every answer demonstrates the opposite of what new-space companies want.
- **Security clearance is required for a significant fraction of defence propulsion roles.** Solid rocket motors, liquid engines for tactical missiles, and hypersonic propulsion are clearance-gated (US TS/SCI, UK DV, Australian NV2/PV). Engineers introducing avoidable friction regarding their clearance eligibility are at a disadvantage.
- **The history of the field is technical content, not background.** The Pogo instability in the Saturn V, the SSME blade fractures, the Space Shuttle SRB O-ring analysis — these are case studies in what the hardware does when analysis is wrong. Candidates who treat this history as engineering content demonstrate necessary intellectual engagement.

PART II · 20 SPECIFIC PREPARATION TASKS

Intern / Graduate

1. Derive and apply the rocket engine performance equations — I_{sp} , c^* , C_f , and the Tsiolkovsky Rocket Equation 6–8 h

Skill: The foundational analytical framework of rocket propulsion

Objective: The performance of a rocket engine is fully characterised by three parameters: characteristic velocity c^* (which depends only on the chamber conditions), the thrust coefficient C_f (which depends on the nozzle area ratio and ambient pressure), and specific impulse $I_{sp} = (c^* \times C_f)/g_0$. A candidate who can derive these from conservation equations and compute approximate values without a table demonstrates expected quantitative fluency. The Tsiolkovsky Rocket Equation ($\Delta v = I_{sp} \times g_0 \times \ln(m_0/m_f)$) connects engine performance to mission performance.

Done Signal: You can derive the thrust equation from first principles, derive the expression for I_{sp} from nozzle exit velocity, and compute approximate I_{sp} values for LOX/RP-1 and LOX/LH₂ at representative chamber pressures. You can apply the Tsiolkovsky Equation to estimate the propellant mass fraction required for a given Δv , and explain why stage separation improves overall Δv .

2. Understand propellant combinations — performance, density, storability, and the reasoning behind common choices 4–5 h

Skill: Propellant literacy as the starting point of every engine design

Objective: The choice of propellant determines the achievable specific impulse, bulk density (density impulse), handling infrastructure, and material compatibility. A graduate who cannot explain why Falcon 9 uses LOX/RP-1 for its first stage but hydrogen was used for the Space Shuttle, or why storable hypergolics remain the choice for spacecraft apogee engines despite toxicity, is missing vital propellant reasoning.

Done Signal: For LOX/LH₂, LOX/RP-1, LOX/CH₄, NTO/MMH, and solid propellants (HTPB/AP/Al), you can state: approximate vacuum I_{sp} , storability, ignition method, and primary application rationale. You can explain why methane has emerged as the preferred propellant for reusable first stages, and why hydrazine is being phased out of new designs.

3. Understand engine cycles — gas generator, staged combustion, expander, and pressure-fed — at engineering depth 4–5 h

Skill: The thermodynamic architecture determining performance, complexity, and failure modes

Objective: Each cycle has a characteristic performance level, complexity profile, and set of failure modes. Staged combustion recovers performance at the cost of substantially higher complexity. The expander cycle is elegant and inherently stable but limited in thrust. Understanding why SpaceX chose full-flow staged combustion for Raptor, or why pressure-fed engines remain in use for upper stages, are practical judgements engineers make daily.

Done Signal: For gas generator, staged combustion (fuel-rich and oxidiser-rich), full-flow staged combustion, expander, and pressure-fed cycles: you can draw the propellant flow schematic, state the primary performance advantage and engineering challenge, name one operational engine of each type, and explain the associated failure mode.

4. Prepare behavioural answers that connect engineering rigour to hardware outcome 2–3 h

Skill: STAR-format stories with the test-failure-learning emphasis propulsion employers specifically probe

Objective: Employers specifically probe how candidates respond to hardware that behaves unexpectedly and anomalies in test data. Weak stories end with “the test went as planned.” Strong stories end with identifying anomalous oscillations, tracing them to specific manifold resonances, and correcting them before flight qualification. Hardware surprises are the curriculum of propulsion engineering.

Done Signal: You have four STAR stories covering: a hardware or test anomaly you identified and investigated; a time you made a technical decision under time pressure; a cross-disciplinary collaboration; and a failure that changed your approach to a subsequent problem. At least two stories reference specific hardware behaviour and diagnostic processes.

5. Work through a nozzle design — area ratio, contour, and the effect of ambient pressure on performance 4–5 h

Skill: The conversion of chemical energy into directed thrust — the nozzle as the performance-delivery mechanism

Objective: The nozzle area ratio determines how completely combustion gases expand. For sea-level operation, the optimum area ratio is lower than for vacuum operation. Understanding why a

large area ratio improves vacuum performance but causes flow separation at sea level, and how the Prandtl-Meyer expansion process determines the exit velocity distribution, are daily nozzle physics applications. 4–5 h

Done Signal: You can compute the optimal exit area ratio for a specified chamber pressure, propellant combination, and ambient pressure using isentropic flow relations. You can explain why over-expanded nozzles produce flow separation and the practical consequence. You know what the Mach number at the nozzle throat is and why a truncated ideal contour (TIC) is typically used.

6. Get hands-on with a chemical equilibrium or rocket performance code — CEA, RPA, or equivalent 4–6 h

Skill: Computational tool fluency as the baseline of propulsion performance analysis

Objective: NASA's CEA code is the canonical tool for computing rocket engine performance. A candidate who has used CEA or RPA to map engine performance across a range of mixture ratios and chamber pressures — and who understands why the mixture ratio for maximum I_{sp} differs from maximum density impulse — has practical engagement with propulsion analysis that textbook exercises cannot substitute.

Done Signal: You have used CEA or RPA to compute the performance of at least two propellant combinations (LOX/RP-1 and LOX/LH₂) across a range of mixture ratios, chamber pressures, and area ratios. You can explain why the maximum I_{sp} does not occur at the stoichiometric mixture ratio and interpret a performance map.

Mid-Level (~3–8 years)

All graduate tasks still apply. Behavioural stories must come from professional or serious academic contexts involving real hardware, real test data, or real design decisions with documented consequences.

7. Build a quantitative understanding of combustion instability — mechanisms, characterisation, and suppression 5–7 h

Skill: The most technically demanding and safety-critical engineering challenge in liquid rocket engines

Objective: High-frequency combustion instability remains the primary technical risk in high-performance liquid engine development. It involves coupling between acoustic modes and combustion heat release. If the heat release perturbation is sufficiently in phase with acoustic pressure oscillation (Rayleigh criterion), it amplifies and destroys the chamber. Engineering responses involve injector design, baffles, and Helmholtz resonators.

Done Signal: You can compute the first tangential, longitudinal, and radial acoustic mode frequencies of a cylindrical chamber, explaining why tangential modes are most destructive. You can state the Rayleigh criterion and what a combustion bomb test reveals. You can name two injector design features that reduce instability susceptibility.

8. Understand turbopump design and failure modes — pump performance, NPSH, cavitation, and rotordynamics 5–6 h

Skill: The highest-risk component in a liquid propellant engine, at engineering depth

Objective: The turbopump operates at extreme speeds and temperatures, pumping propellants from tank pressure to chamber pressure. The key engineering challenges are Net Positive Suction Head (NPSH) requirements, cavitation (vapour bubbles causing structural damage), rotordynamic stability (whirl instability), and seal design. Designing or analysing a turbopump demonstrates hard-to-fake engineering depth.

5–6 h

Done Signal: You can explain what NPSH is physically and how it is reduced through inducer design. You can describe the mechanism of cavitation and its effect on pump performance curves. You can explain rotordynamic whirl and name at least one turbopump failure from the history of rocket propulsion, describing the investigation's findings.

9. Understand regenerative cooling — heat flux, coolant flow design, and failure mode analysis

4–6 h

Skill: The thermal protection methodology for high-performance liquid engines

Objective: Heat fluxes at the nozzle throat can exceed 100 MW/m^2 . Regenerative cooling routes propellants through channels in the chamber wall before injection. Designing this requires computing local heat flux distributions, coolant flow rates, wall temperatures, and thermal stress. The burnthrough failure mode is a common cause of test failures.

Done Signal: You can describe the one-dimensional analysis procedure for regenerative cooling: computing hot-gas heat flux (Bartz correlation), specifying coolant channel geometry, computing heat transfer coefficients, and solving for wall temperature. You can explain what a burnthrough failure mode looks like in test data and how to reduce the risk.

10. Prepare a story about diagnosing a propulsion anomaly using test data — what you saw, what you hypothesised, and what the hardware told you

2 h

Skill: Test data interpretation as applied engineering judgement under operational pressure

Objective: Every hot fire test produces data that tells engineers what is happening inside the engine. Distinguishing a real combustion event from an instrumentation anomaly, identifying cavitation onset, or recognising a combustion instability frequency signature are skills developed only from real test data exposure.

Done Signal: You have a story about a test anomaly where data showed something unexpected, hypotheses were formed, and a root cause was diagnosed. The story names the engine, test campaign, specific anomaly in the data (pressure trace, temperature record), diagnostic analysis performed, and the specific corrective design change implemented.

11. Understand solid rocket motor ballistics — burn rate, pressure-time trace, and grain geometry design

4–5 h

Skill: The propulsion methodology for solid rocket boosters, upper stages, and tactical missiles

Objective: Solid motor thrust-time profiles are determined entirely by grain geometry and the burn rate law. Understanding why a star-grain produces a neutral burn, why a cylindrical bore burns regressively, and how the K_n ratio determines chamber pressure through the Vieille law are essential. The stability condition requires the burn rate exponent $n < 1.0$.

Done Signal: You can explain the Vieille burn rate law ($r = a \cdot P^n$) and why $n < 1.0$ is required for stable operation. You can describe how grain geometry (star, cylindrical bore) determines the thrust-time profile. You can compute the expected chamber pressure for a specified burning surface area, throat area, and burn rate.

12. Build working knowledge of ignition systems and startup transients — the highest-risk phase of engine operation 3–4 h

Skill: The engineering of the moment most likely to destroy the engine

Objective: The engine startup transient is highly susceptible to catastrophic hard starts if unburned propellant accumulates before ignition. The sequence of events must be precisely timed. Ignition system design (pyrotechnic, hypergolic slug, torch, spark) determines reliability and propellant compatibility.

Done Signal: You can explain what a hard start is physically and the consequences of propellant accumulation before ignition. You can describe three ignition system types, state their advantages and disadvantages, and describe what the chamber pressure trace looks like during a normal startup versus a hard start.

13. Prepare for a propulsion system sizing problem under interview conditions 5–7 h

Skill: First-principles engine sizing as a timed engineering demonstration

Objective: Mid-level interviews frequently include a live sizing problem (e.g., “size a LOX/LH₂ engine for 100 kN thrust, I_{sp} 450 s, 60 bar, O/F 5.5”). This requires structured application of performance equations, isentropic flow relations, pump work equations, and turbine energy balances with explicit unit tracking under time pressure.

Done Signal: You can perform, in under 60 minutes with a calculator, a complete first-order engine sizing: computing propellant flow rates, throat area, area ratio, exit conditions, pump inlet/discharge conditions, and approximate turbopump power. You know the exact order of steps and common unit conversion errors.

Senior Level (*Principal Engineer, Technical Lead, Chief Engineer*)

All mid-level tasks still apply. Combustion stability knowledge must extend to specific instability characterisation methods. Turbopump knowledge must include experience with a real turbopump development or anomaly investigation.

14. Prepare to walk through a complete engine or motor development programme — from requirements through qualification firing 4–6 h

Skill: End-to-end propulsion system development ownership at the level a senior engineer must demonstrate

Objective: Senior engineers are expected to have taken a propulsion system from performance requirements through integrated testing to flight qualification. The interview probes the specificity of this experience: owning the stability characterisation, identifying tangential mode instabilities, redesigning the injector pattern, and demonstrating margin recovery. Specificity is the signal.

Done Signal: For one propulsion system you personally owned a technical thread for, you can discuss: originating performance requirements, key design trade-offs, test campaign structure, the most significant anomaly discovered, your diagnostic approach, the design change implemented, and the qualification evidence that demonstrated resolution.

15. Prepare a technically grounded position on one active frontier in rocket propulsion — reusability, additive manufacturing, or methane engines 4–5 h

Skill: Forward-looking technical awareness at senior engineering depth

4–5 h

Objective: Reusability requires understanding cyclic thermal fatigue and high-cycle fatigue in turbopump blades. Additive manufacturing enables complex cooling geometries but introduces porosity and inspection challenges. Methane engines face icing issues during propellant loading. Senior engineers must engage at technical depth with these frontiers, identifying unsolved engineering problems.

Done Signal: You have a prepared 3-minute technical discussion covering a specific engineering challenge on an active frontier: material fatigue life prediction for oxygen-rich preburners, qualification methodology for additively manufactured copper-alloy chambers, or methane icing mitigation approaches. The position is technically grounded and connects to your background.

16. Prepare to lead a technical discussion on a propulsion system architecture trade-off for a specific mission scenario

4–5 h

Skill: Architectural decision-making at senior engineering speed under challenge

Objective: Senior engineers advise on architecture decisions impacting mission performance, cost, and schedule. “For a reusable first-stage booster targeting 50 flights, choose your propellant and cycle and justify it” requires reasoning through coking, relight reliability, staged combustion complexity versus gas generator simplicity, and density impulse. You must structure this reasoning live.

Done Signal: You have worked through three architecture trade-off problems (e.g., reusable booster propulsion, in-space cislunar tug, upper stage propulsion). You can state the constraining requirements, two viable options with principal trade-offs, your recommended architecture, and the specific risk that would cause you to reconsider.

17. Understand the propulsion system test architecture — from component testing through system-level hot fire to flight qualification

3–4 h

Skill: The test-driven development methodology that is the defining characteristic of propulsion engineering

Objective: The test campaign architecture — from injector element tests and turbopump hydraulic tests to full engine hot fires — is the engineering product senior engineers design. It must reveal failure modes before flight and characterise the operating envelope. Senior engineers must explain why a test sequence was chosen and how anomalies changed the approach.

Done Signal: For one propulsion programme you have worked on, you can describe the test campaign architecture, the specific technical risk each phase was designed to retire, and what actually happened. You can explain “risk retirement” and describe what a qualification firing requirement typically involves (operating time, off-nominal conditions).

18. Build quantitative estimation fluency for propulsion system performance — without a simulation

4–5 h

Skill: Senior-speed first-principles reasoning about engine and mission performance

Objective: Senior engineers sanity-check performance claims and evaluate engine concepts in real time. Estimating propellant mass fractions, turbopump power for a specified chamber pressure, throat area for a specified thrust, or steady-state heat fluxes requires performance equations and physical intuition, not simulations.

Done Signal: You have worked through five estimation problems out loud: propellant mass fraction

from the Rocket Equation, turbopump power from propellant flow and pressure rise, throat area from thrust/chamber conditions, heat flux at the nozzle throat, and solid motor burn time. You arrive at order-of-magnitude answers with stated uncertainty. 4–5 h

All Levels

19. Prepare a specific answer to “Why propulsion — and why this specific technology area?” that reflects genuine technical engagement 1–2 h

Skill: Technically specific motivation that distinguishes propulsion-committed candidates from aerospace generalists

Objective: Employers want engineers with deep technical opinions: why full-flow staged combustion creates unique challenges, why additive manufacturing changes cooling channel designs, or why methane is preferred over RP-1 for reusability. A generic “I like rockets” answer falls flat compared to engaging with a specific technical challenge in the modern landscape.

Done Signal: You have a 90-second answer naming your target propulsion area (liquid bipropellant, solid motors, electric propulsion, hypersonic), identifying one specific active engineering challenge, explaining what makes it technically hard, and connecting your background to its solution.

20. Map your technical background to a specific employer type, propulsion technology, and role — and verify security clearance eligibility 1–2 h

Skill: Precise self-positioning in an employer landscape spanning new-space startups to classified defence programmes

Objective: Working environments vary wildly. A new-space startup engineer owns an entire turbopump and tests changes in weeks. A prime contractor engineer owns one subsystem within a long change management process. Solid motor engineers at legacy primes may require SECRET clearances. Failing to differentiate these environments signals poor fit.

Done Signal: You can state in under 60 seconds: the employer type you are targeting (new-space startup, legacy prime, spacecraft thruster manufacturer), the specific propulsion technology most relevant to your background, the analysis tools you use fluently, your security clearance status/eligibility, and two specific technical contributions your background brings to the target role.

SUMMARY REFERENCE TABLE

#	Task Description	Target Level	Effort
1	Derive Isp, C*, Cf, and Tsiolkovsky — from first principles, with numbers	All	6–8 h
2	Propellant combinations — performance, density, storability, selection reasoning	All	4–5 h
3	Engine cycles — GG, staged combustion, expander, pressure-fed — schematics	All	4–5 h
4	STAR stories — hardware anomaly, time-pressure decision, cross-discipline	All	2–3 h
5	Nozzle design — area ratio, bell contour, flow separation, performance penalty	All	4–5 h
6	CEA or RPA hands-on — performance maps, optimal O/F ratio, area ratio trade	All	4–6 h
7	Combustion instability — Rayleigh criterion, acoustic modes, bomb test	Mid+	5–7 h
8	Turbopump design — NPSH, cavitation, rotordynamics, historical failure	Mid+	5–6 h
9	Regenerative cooling — heat flux, channel design, burnthrough failure mode	Mid+	4–6 h
10	Test anomaly diagnosis story — data-driven, specific hardware, root cause	Mid+	2 h
11	Solid motor ballistics — Vieille law, stability condition, grain geometry	Mid+	4–5 h
12	Ignition systems and startup transients — hard start, igniter types, pressure	Mid+	3–4 h
13	Engine sizing problem under time pressure — flow rates, throat, turbopump	Mid+	5–7 h
14	Complete development programme story — requirements through qualification firing	Senior	4–6 h
15	One active frontier — reusability, additive manufacturing, or methane engines	Senior	4–5 h
16	Propulsion architecture trade-off — live, under challenge, specific	Senior	4–5 h
17	Test campaign architecture — phase structure, risk retirement, qualification	Senior	3–4 h
18	Quantitative estimation — five problems, out loud, assumptions before numbers	Senior	4–5 h
19	“Why propulsion — why this technology area?” — technically specific	All	1–2 h
20	Employer type / technology / clearance self-positioning	All	1–2 h

THE STRUCTURAL REALITY

Rocket propulsion is the only engineering discipline where the system routinely operates at conditions no material was designed to withstand — and where the margin between designed operation and catastrophic failure is measured in milliseconds and in fractions of a percent of chamber pressure. A turbopump bearing that fails at 50,000 rpm does not gracefully degrade. A combustion instability that becomes established in a 500 bar chamber does not stabilise at a lower amplitude. A nozzle throat that burns through during a qualification test does not produce a partial failure. The engineering culture this creates — the obsession with margins, with test data interpretation, with failure mode identification before the hardware reveals them — is the defining characteristic of the best propulsion engineers, and it is what every technical interview in the field is ultimately assessing.

The field is simultaneously the most quantitative and the most empirical in aerospace engineering. The Rocket Equation is exact. The thrust coefficient derivation is closed-form. The isentropic flow relations are the same equations that have been in textbooks since the 1940s. And yet, the F-1 programme required 2,000 full-scale injector tests to solve its combustion instability problem — because the physics of propellant atomisation, mixing, and combustion in a complex injector pattern operating at 70 bar and 3,500 K is not, today or then, reducible to first principles. The quantitative framework tells you what the engine should do. The test campaign tells you what the hardware actually does. The propulsion engineer's job is to close the gap between them, one test at a time, until the hardware agrees with the analysis — and to understand, when it does not, exactly why.

Engineers who succeed in propulsion are those who hold both of these realities simultaneously: the mathematical precision of the performance equations and the stubborn, often violent, unpredictability of real combustion hardware. They use the analysis to design, and the test data to learn, and they approach the gap between them not as evidence of inadequate analysis but as the most interesting engineering problem the discipline has to offer — because it always is.